

ABHINAV AVASARALA

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Education

North Carolina State University (NCSU)

May 2027 (Expected)

B.S. Computer Engineering

Raleigh, NC

- **GPA:** 4.00/4.00
- **Courses:** Software Development, Statistics, Data Structures & Algorithms, C & Software Tools, Statistics for Engineers, Operating Systems, Signals & Circuits, Machine Learning, Digital Design, Architecture of Parallel Computers

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL, Verilog, x86 Assembly
Agentic AI & LLMs: LangGraph, MCP, RAG, Multi-Agent Orchestration, Prompt Engineering, Cursor
Cloud & Infrastructure: GCP Associate Cloud Engineer, Docker, Kubernetes, Prometheus
GPU / AI Systems: CUDA, cuBLAS, Triton, PyTorch, Nsight Systems, Nsight Compute
Software & Tools: FastAPI, Flask, Apache Kafka, PostgreSQL, ReactJS, Git, NumPy, Pandas

Experience

iQuadra Information Services | ReactJS, Axios, REST, Figma

May 2025 – August 2025

Software Engineer Intern

Remote

- Developed a bug-tracking system using ReactJS front-end, Node + Express back-end, PostgreSQL database, and Axios integration
- Designed **12+ RESTful API endpoints** using Node + Express with PostgreSQL for persistence
- Improved bug resolution time by **~20%** through priority-tagging and streamlined workflow
- Led a team of 3 front-end devs to build responsive UIs from Figma designs using React

Projects

KubeMind | Python, LangGraph, GPT-4o, NeMo Guardrails, GCP, GKE, Prometheus, Kubernetes

- Built a natural language Kubernetes CLI agent using **LangGraph** and **GPT-4o** that automates node pool provisioning, deployment scaling, and cluster diagnostics via single English commands
- Implemented a 3-layer safety system with **NeMo Guardrails** and a deterministic risk classifier to intercept dangerous LLM-generated ops before execution; enforces name-typed confirmation for irreversible mutations
- Designed an **agent evaluation harness** benchmarking GPT-4o vs. Claude across 19 labeled test cases with **precision/recall/F1 scoring**; achieved 89% pass rate and surfaced model-specific planning failures
- Integrated **Prometheus** on **GKE** for natural language health queries flagging crash-loops, CPU spikes, and anomalies as pass/warn/critical; built deterministic undo across 7 op types using state snapshots

Paper2Proto - Hackathon Project | Python, FastAPI, LangGraph, AWS Bedrock, OpenSearch, S3

- Designed a **4-agent orchestration system** using LangGraph to convert research papers into structured implementation plans, reducing manual analysis time by **60%**
- Built an end-to-end RAG pipeline processing 10–20 page PDFs (5,000+ tokens) per upload, automating document ingestion from S3, chunking, embedding, and vector indexing in OpenSearch using AWS Bedrock Knowledge Base
- Implemented session-scoped metadata filtering to ensure **100% context isolation** across user uploads while retrieving top-*k* relevant chunks in **<500ms** average latency

CommandCopilot | Python, Apache Kafka, llama-cpp-python, NCSU HPC

- Built a distributed terminal assistant across 3 NCSU HPC nodes that diagnoses shell errors in real time and returns actionable fixes in under 5 seconds
- Engineered a two-tier inference engine with 25+ regex rules for sub-1s fixes and on-cluster Phi-2 2.7B LLM fallback over Kafka messaging, achieving a 5x latency improvement vs external API calls
- Quantized Phi-2 2.7B to GGUF Q4 via llama-cpp-python for CPU-only inference, reducing model memory footprint by ~75% vs full precision
- Designed for horizontal scalability using Kafka consumer groups and partition-based load distribution, allowing throughput to scale linearly by adding consumer instances per partition

Leadership / Extracurricular

- **IBM-NCSU Pathfinder program:** Learned about various IBM & Red Hat technologies like Ansible & watsonx
- **180 Degrees Consulting:** Led a team of 4 student consultants to improve a non-profit client's marketing strategy